FINC5001

Module 7

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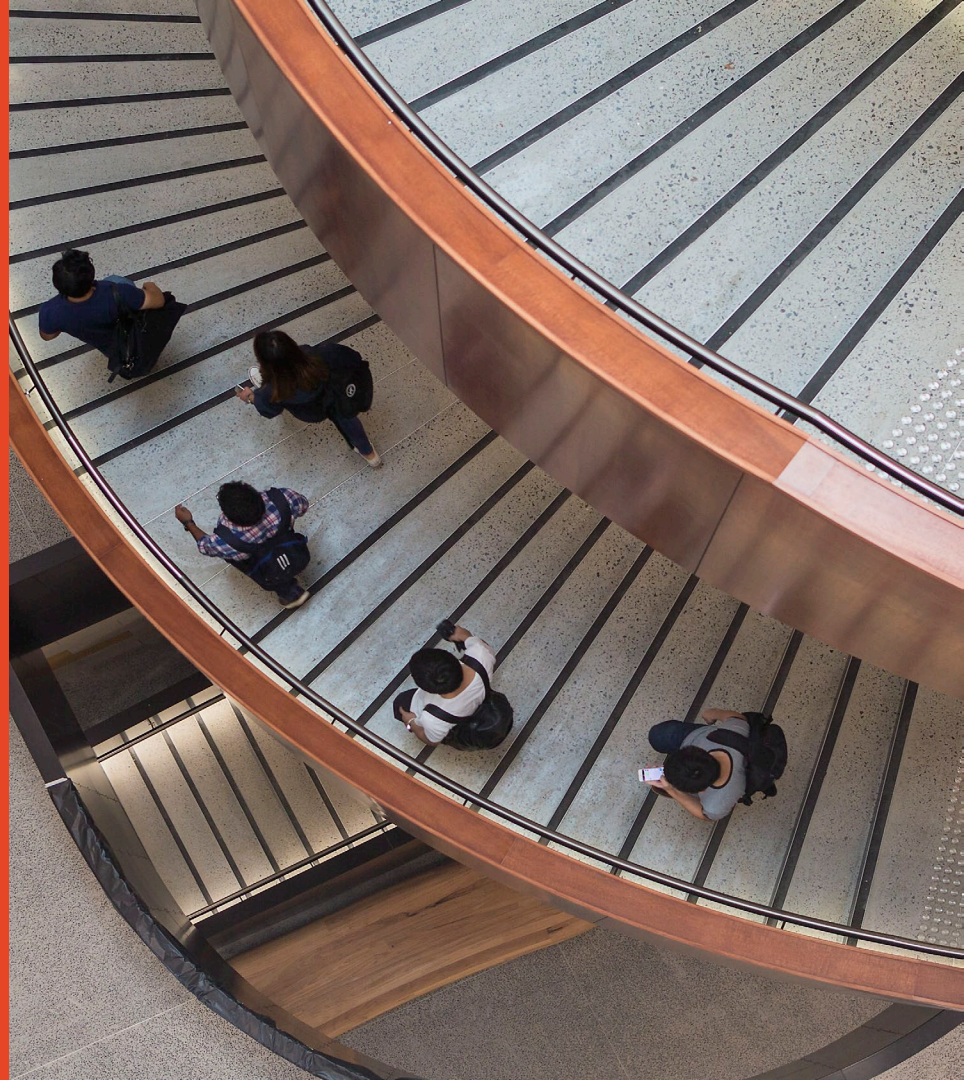


Lecture Outline

This lecture will discuss:

- The company cost of capital, with regard to:
  - Calculation of the company cost of capital
  - Estimation of the cost of debt capital
  - Estimation of the cost of equity capital
  - The impact of corporate tax on the company cost of capital
- The calculation and forecasting of free cash flows, including:
  - Cash flow reporting in financial statements
  - Preparation of free cash flow forecasts

Value of a firm

$$V = \sum_{t=1}^{00} \frac{F_t}{(1+r)^t}$$

where :

F_t = net cashflows after tax but before interest

r = after tax required rate of return (company cost of capital)

The company cost of capital

Shareholders

$10 million invested

Opportunity cost = 16%

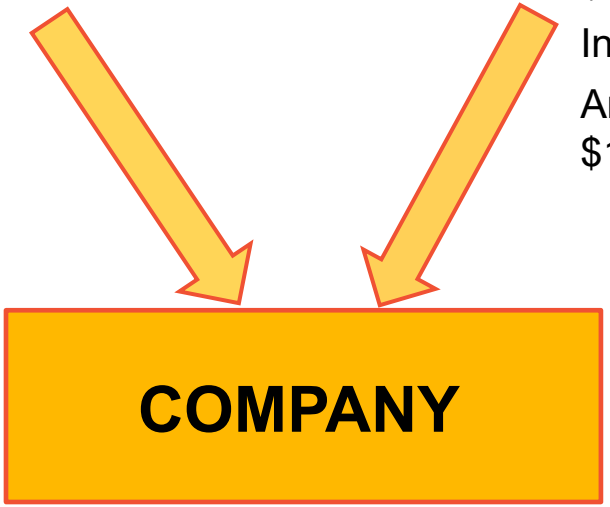
Annual cost = $1.6 million

Debtholders

$10 million invested

Interest rate = 10%

Annual cost = $1million



What is the cost of capital?

Capital	Amount	Annual Cost	% of Total	Annual Cost
Equity	$10m	16%	50%	$1.6m
Debt	$10m	10%	50%	$1.0m
Total	$20m	?	100%	$2.6m

Total required return

- Required return/invested capital
  - $2.6m/$20m = 13%

Total required return (cont'd)

– Or calculate using the Weighted Average Cost of Capital (WACC), with no tax, as follows:

$$WACC = \left(k_d \times \frac{D}{D+E}\right) + \left(k_e \times \frac{E}{D+E}\right)$$

where:

D = market value of debt
E = market value of equity
k_d = cost of debt
k_e = cost of equity

Example: WACC

- *The market value of debt and equity of a company are $10 million each. The debtholders of the company charge an interest rate of 10% p.a. while the equity holders require a return of 16% on their investment. What is the company cost of capital?*

$$
\begin{aligned}
WACC &= \left(k_d \times \frac{D}{D+E}\right) + \left(k_e \times \frac{E}{D+E}\right) \\
&= \left(0.10 \times \frac{10}{10+10}\right) + \left(0.16 \times \frac{10}{10+10}\right) \\
&= (0.10 \times 0.50) + (0.16 \times 0.50) = 0.13 = 13\%\ p.a.
\end{aligned}
$$

The cost of debt capital

– In topic 4, the following expression was used to find the value of a debt security:

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n}$$

where:

D = the market value of the debt

n = the time to maturity of the debt

C = the dollar interest paid on the debt

F = the face value of the debt

r_d = the discount rate **= THE BEFORE TAX COST OF DEBT**

Estimating the cost of debt

- There are several approaches:
  - Assume the debt is risk-free, and use the 10-year government bond yield
  - Add 100 – 200 basis points (1 – 2%) to the 10-year bond yield to allow for risk
  - Use the following equation:

$$k_d = \frac{\text{net interest}}{\text{average net debt}}$$

where:

net interest = interest paid - interest received

average net debt = reported book value of debt - cash

The after tax cost of debt

- Because interest expenses are tax deductable we often want the after tax cost of debt:

$$= r_d\,(1-t_c)$$

where:
r_d = required return on debt
t_c = the corporate tax rate

eg. if r_d = 10%; t_c = 30% then
after tax cost of debt = 10% (1-.30) = 7.0%

Total required return (cont'd)

– Therefore the after-tax Weighted Average Cost of Capital (WACC) equals:

$$WACC = \left(k_d \times (1-T) \times \frac{D}{D+E} \right) + \left(k_e \times \frac{E}{D+E} \right)$$

where:

D = market value of debt
E = market value of equity
k_d = cost of debt
k_e = cost of equity

The cost of preference shares

– The **cost of preference shares** is the rate of return investors require of the firm when they purchase its preference shares.

$$r_{ps} = \frac{d}{P_0}$$

where:

P_0 = current value of share

d = (constant) dividend per period

r_{ps} = required return on the preference share

Cost of preference shares example

- Consider the preference shares of Adelaide Power Company, which are trading at \$23.35 per share. What will be the required rate of return on these shares if they have a par value of \$25 and pay an annual dividend of 5.3%?

- Annual dividend = 0.053 x \$25 = \$1.325

- r_{ps} = \$1.325 ÷ \$23.35 = **.0567 or 5.67%**

Cost of ordinary shares

- The minimum required rate of return on shares for the risk involved
- Estimation methods:
  - implied from current prices
  - modelled using the CAPM

Implying the cost of ordinary shares

$$P = \sum_{t=1}^{00} \frac{d_t}{(1+r_e)^t}$$

where:
P = the current price of shares
d_t = the dividend paid in year t
r_e = discount rate of equity of the firm given its risk

Implying the cost of ordinary shares

– The **cost of ordinary shares** is the rate of return investors require of the firm when they purchase its ordinary shares.

$$P_0 = \frac{d_1}{k_e - g}$$

where:

P_0 = current value of share

d_1 = dividend in period one

k_e = cost of ordinary shares

g = constant growth rate

Implying the cost of ordinary shares

– This can be rearranged to solve for r_e as follows:

$$r_e = \frac{d_1}{P_0} + g$$

where:

P_0	=	current value of share
d_1	=	dividend in period one
r_e	=	required return on the ordinary share
g	=	constant growth rate

Example

A stock is currently trading at $6, its dividends are expected to grow at a rate of 6% beyond FY2023, but are forecast as follows for the next 4 years:

2020	2021	2022	2023
23.3cps	26.1cps	29cps	32.1cps

What is the stocks equity cost of capital?

Example - Solution

Cost of capital is the r_e that makes the following equation true:

$$\$6.00 = \frac{\$0.233}{(1+r_e)^1} + \frac{\$0.261}{(1+r_e)^2} + \frac{\$0.29}{(1+r_e)^3} + \frac{\$0.321}{(1+r_e)^4} + \frac{\dfrac{\$0.321 \times (1.06)}{r_e - 0.06}}{(1+r_e)^4}$$

r_e = 10.45%

Estimating the required return on ordinary shares using the CAPM

$$(k_e) = r_f + \beta_e \times E(r_m - r_f)$$

where:

k_e = required return on equity for the stock

r_f = current risk free rate of return

$E(r_m - r_f)$ = market risk premium

β_e = regression co-efficient estimated from historical data for the stock

Estimating the cost of ordinary shares using CAPM

- Using CAPM to estimate the cost of equity:
  1. Estimate the beta of the stock using historical data
  2. Determine the current risk-free rate
  3. Determine the expected market risk premium
  4. Substitute these values into the CAPM equation

Estimating market risk premium

- There are 3 approaches to estimating market risk premium:
  - Use historical average returns on the stock market
  - Imply from share prices and analysts' forecasts
  - Estimate the relationship between stock market volatility and stock market returns

Estimating market risk premium (cont)

1. Using historical average returns on the stock market

Assumptions:

- Realised returns are equivalent to the returns expected by market participants for bearing risk in the future
- The market risk premium is constant over time

Estimating market risk premium (cont)

2. Using analysts' forecasts of earnings and dividends:

$$r_{ei} = \frac{d_{1i}}{P_{it}} + g_i$$

where:

r_{ei} = the equity cost of capital for stock *i*

d_1 = the one-year-ahead forecast dividend for stock *i*

g_i = the constant growth rate for stock *i*

P_{it} = the current price of stock *i*

Estimating market risk premium (cont)

3. Estimating the relationship between stock market volatility and stock market returns

$$r_{m,t} - r_{f,t} = a + b\sigma^2(t)$$

where

$r_{m,t}$	=	the market return over period *t*
$r_{f,t}$	=	the risk-free rate of return over period *t*
b	=	reward-to-risk ratio
σ2(t)	=	stock return volatility period *t*

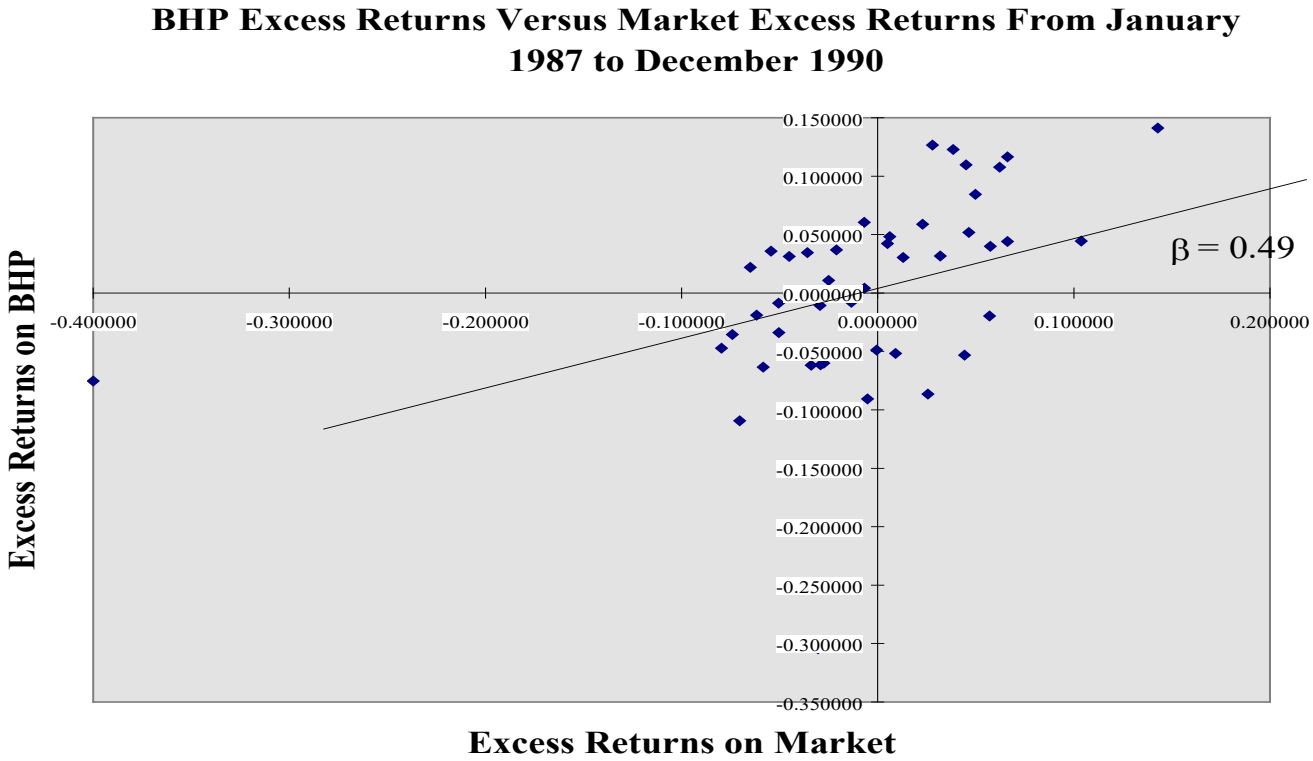
Problems with using the CAPM to estimate the cost of ordinary shares

Stability of β 's over time

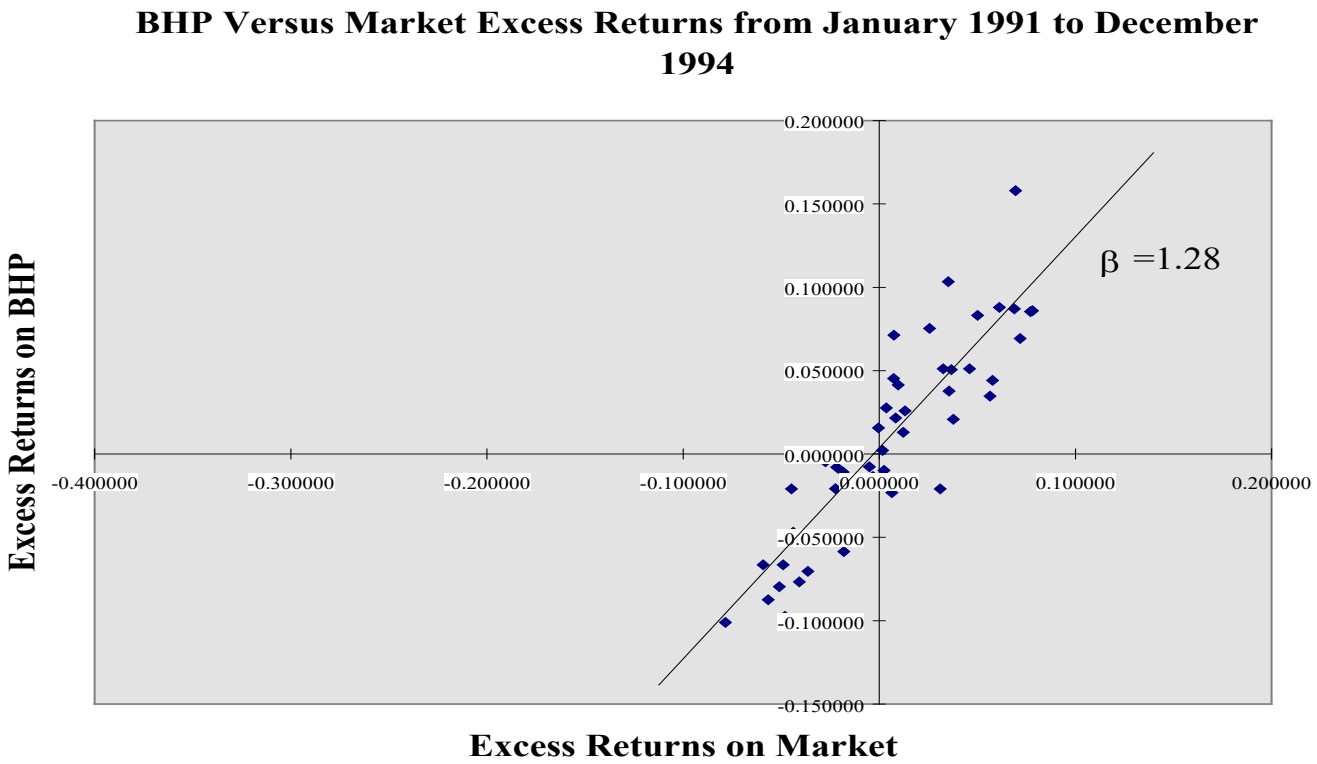
1. Stocks may change in risk through time
2. β estimates are from a limited number of observations -> measurement error

eg. BHP β based on different time periods

Problems with using the CAPM to estimate the cost of ordinary shares



Problems with using the CAPM to estimate the cost of ordinary shares



Example- estimating the cost of ordinary shares

- *Burswood Ltd has a beta of 1.31, the market risk premium is 5.3% and the risk free rate is currently 5.5%. What is the cost of equity capital?*

$$k_e = r_f + \beta_i \cdot E(r_m - r_f)$$
$$k_e = 0.055 + 1.31\ (0.053) = 0.1244$$

A digression: *the imputation system of tax*

- For every $1 of tax paid by the company, the shareholder gets a franking credit (tax credit)
- Thus $1 of corporate income only gets taxed once, at the personal tax rate of the shareholder

Example: the impact of imputation on shareholder returns

A company's income of $1 is paid out as a dividend. What is the shareholder's after-tax income if his tax rate is 45%?

		Classical	Imputation
Corporate level	Net income	$1.00	$1.00
	Company tax	0.30	0.30
	Cash dividend	0.70	0.70
Shareholder level	Taxable income	0.70	1.00
	Personal tax liability	0.32	0.45
	Franking credit	n.a.	0.30
	Tax paid	0.32	0.15
	After-tax income	0.38	0.55

Returns to shareholder under imputation

- *Under the classical system of tax:*

$$r_t = [(p_t - p_{t-1}) + d_t]/p_{t-1}$$

- *Under imputation (assuming shareholders use all franking credits)*

$$*r_t = [(p_t - p_{t-1}) + (d_t + C_t)]/p_{t-1}$$

where

$$d_t + C_t = \text{grossed up value of dividends} = d_t/(1-t_c)$$

CAPM under an imputation system of tax

– The CAPM also needs to be redefined so that the value of franking credits is included in returns on both the stock and on the market:

$$r_j' = r_{ft} + \left[E\left(r_{mt} + \tau_{mt}\right) - r_{ft}\right] \times \beta_j$$

where:

r_j' = expected return on stock *j* before personal tax, including the value of franking credits used

r_{ft} = risk-free rate of return over period *t*

r_{mt} = return on the market over period *t*

= value of franking credits paid on stocks in the index

β_j = beta of stock *j*

Incorporating the effect of tax into the company cost of capital

- The WACC needs to be adjusted to reflect the effect of tax
- Under the classical system
  - if discounting forecast cash flows estimated *before* tax:

$$WACC = k_d(D/V) + (k_e/[1-t_c])(E/V)$$

  - if discounting forecast cash flows estimated *after* tax:

$$WACC' = (k_d[1-t_c])(D/V) + k_e(E/V)$$

where: t_c = the corporate tax rate

Incorporating the effect of tax into the company cost of capital

- Under the imputation system
  - if discounting forecast cash flows estimated *before* tax:

$$WACC = k_d(D/V) + [k_e/(1 - t_c[1-\gamma])](E/V)$$

  - if discounting forecast cash flows estimated *after* tax:

$$WACC = k_e \times \frac{(1-T)}{(1-T[1-\gamma])} \times \frac{E}{V} + k_d \times (1-T) \times \frac{D}{V}$$

where:

r_e, r_d	=	expected return on equity and debt before tax
E, D, V	=	market value of equity, debt & the firm, respectively
T	=	effective rate of company tax on company cash flow
Y	=	proportion of franking credits valued by the company's shareholders

Incorporating the effect of tax into the company cost of capital (cont)

Example

Using the following information for Tabcorp Holdings Ltd estimate the company cost of capital

Estimated required rate of return on equity = 15.5%

Estimated cost of debt = 8.11% Net debt = $1729.9m

Number of shares = 688 million Share price= $7.72

1) Market value of equity = 688m x 7.72 = $5311.36m
2) Market value of firm = 5311.36m + 1729.9m = $7041.26m

$$r_i = r_E \times \frac{(1-T)}{(1-T[1-\gamma])} \times \frac{E}{V} + r_D \times (1-T) \times \frac{D}{\mathrm{V}}$$

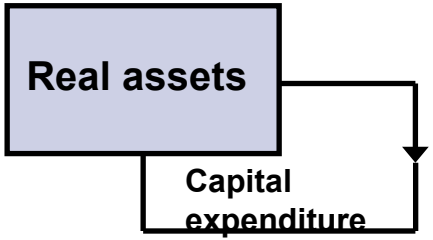
$$= 0.155 \times \frac{(1-0.30)}{(1-0.30[1-1])} \times \frac{5311.36}{7041.26} + 0.0811 \times (1-0.30) \times \frac{1729.9}{7041.26}$$

$$= 0.0958 = 9.58\% \, p.a.$$

Financial reporting and calculating free cash flows

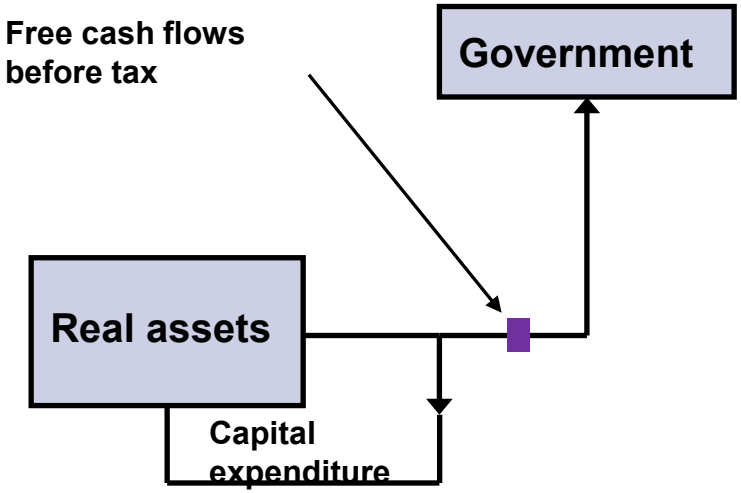
- The cash flows of a company after reinvestment costs (capital expenditure) are distributed to the government, shareholders and debtholders of the company
- The value of a company is determined by the cash flows distributed to its shareholders and debtholders – the free cash flows after tax
- Hence, the free cash flows need to be forecast and discounted at the company cost of capital

Australian corporate financial reporting



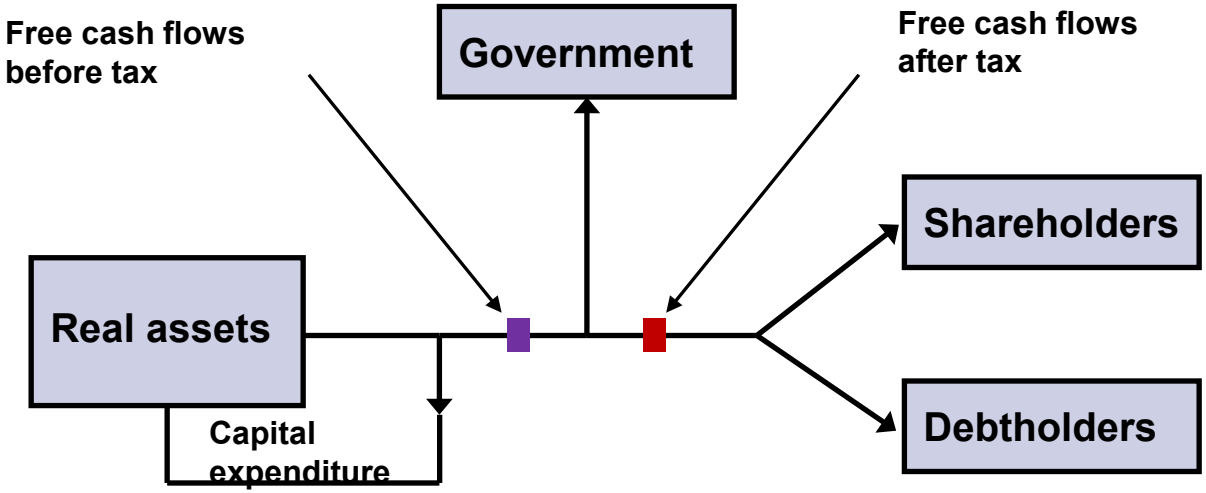
– The real assets of a firm generate cash flows, some of which are reinvested in real assets

Australian corporate financial reporting (cont)



– Some of the free cash flows are then distributed to the government by way of tax

Australian corporate financial reporting (cont)



– Free cash flows after tax are then distributed to capital providers (shareholders and debtholders)

Statement of cash flows

– The cash flow statement is divided into three sections:

Cash flows from operating activities

Cash flows used in investment activities

Cash flows used in financing activities

Statement of cash flows (cont)

– The cash flow statement is divided into three sections:

Cash flows from operating activities

Cash flows used in investment activities

Cash flows used in financing activities

Examples:

- **Cash receipts and payments in the course of operations**
- **Income tax paid**
- **Dividends received**
- **Interest received or paid**

Statement of cash flows (cont)

– The cash flow statement is divided into three sections:

Cash flows from operating activities

Cash flows used in investment activities

Cash flows used in financing activities

Examples:

- **Purchases of property, plant and equipment**
- **Sales of property, plant and equipment**
- **Purchases and sales of businesses**

Statement of cash flows (cont)

– The cash flow statement is divided into three sections:

Cash flows from operating activities

Cash flows used in investment activities

Cash flows used in financing activities

Examples:

- **Funds lent, borrowed or repaid**
- **Proceeds of share issues**
- **Funds used for share buybacks**
- **Dividends paid**

Calculation of free cash flows

- Hence, the free cash flows can be calculated from a company's cash flow statement using the following calculation:

	Cash receipts from operations
Less:	Cash payments from operations
Less:	Net capital expenditure
Equals:	Free cash flows before tax

- The corporate tax rate is then applied to free cash flows before tax to find free cash flows after tax
- Note that all other items on the cash flow statement relate to distribution of cash flows to owners and are irrelevant to valuing the firm

Calculation of free cash flows (cont)

Example

Estimate the value of each Tabcorp Holdings Ltd share, given:

Estimated *real* after-tax company cost of capital = 6.64%

Forecast *real* economic growth = 3.75% FCF before tax = $623.2m

Number of shares = 688 million Net debt = $1729.9m

- (i) Free cash flows after tax: $F_0 = \text{FCF before tax} \times (1 - t_c)$
  $= 623.2 \times (1 - 0.30) = 436.2$
- (ii) Value of firm using constant dividend growth model:

$$V = \frac{F_t(1+g)}{WACC - g} = \frac{436.2(1.0375)}{0.0664 - 0.0375} = \$15659.4$$

Calculation of free cash flows (cont)

Example

Estimate the value of each Tabcorp Holdings Ltd share, given:

Estimated *real* after-tax company cost of capital = 6.64%

Forecast *real* economic growth = 3.75%

Number of shares = 688 million Net debt = $1729.9m

- (iii) Value of equity = value of firm – net book value of debt
- = $15659.4 - $1729.9 = $13929.5
- (iv) Value of each share = value of equity / number of shares
- = $13929.5 /688 = $20.25

Estimating project cost of capital

– Should the firm's WACC be used to evaluate all new investment proposals?

Estimating divisional WACCs

- If a firm undertakes investments with very different risk characteristics, it will try to estimate **divisional WACCs**.

- Divisions are generally defined either by geographical regions (e.g. the Asian region) or industry (e.g. a large integrated oil company may have exploration, production and refining divisions).

Using pure-play firms to estimate divisional WACCs

- Here a firm with multiple divisions may identify a comparable firm with only one division (called ‘pure play’ comparison firms or ‘comps’).

- The estimate of a pure-play firm’s cost of capital can then be used as a proxy for that particular division’s cost of capital.

Flotation costs and project NPV

– **Flotation costs** are fees paid to an investment banker and costs incurred when securities are sold at a discount to the current market price

–Because of flotation costs, the firm will have to raise more than the amount it needs to finance the project.

$$\text{Flotation-cost-adjusted initial outlay} = \frac{\text{Financing needed}}{(1 - \text{Percentage flotation cost})}$$

WACC, flotation costs and project NPV example

–If a firm needs $50 million to finance its new project and the flotation cost is expected to be 7%, how much should the firm raise by selling securities?

$$\text{Flotation-cost-adjusted initial outlay} = \frac{\text{Financing needed}}{(1 - \text{Percentage flotation cost})}$$

= $50 million ÷ (1-0.07) = $53.76 million

– Thus the firm will raise $53.76 million, which includes flotation costs of $3.76 million.

The End

Next week we will look at market efficiency and behavioural finance